

My personal experience on the PXR Blind Challenge

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Outline

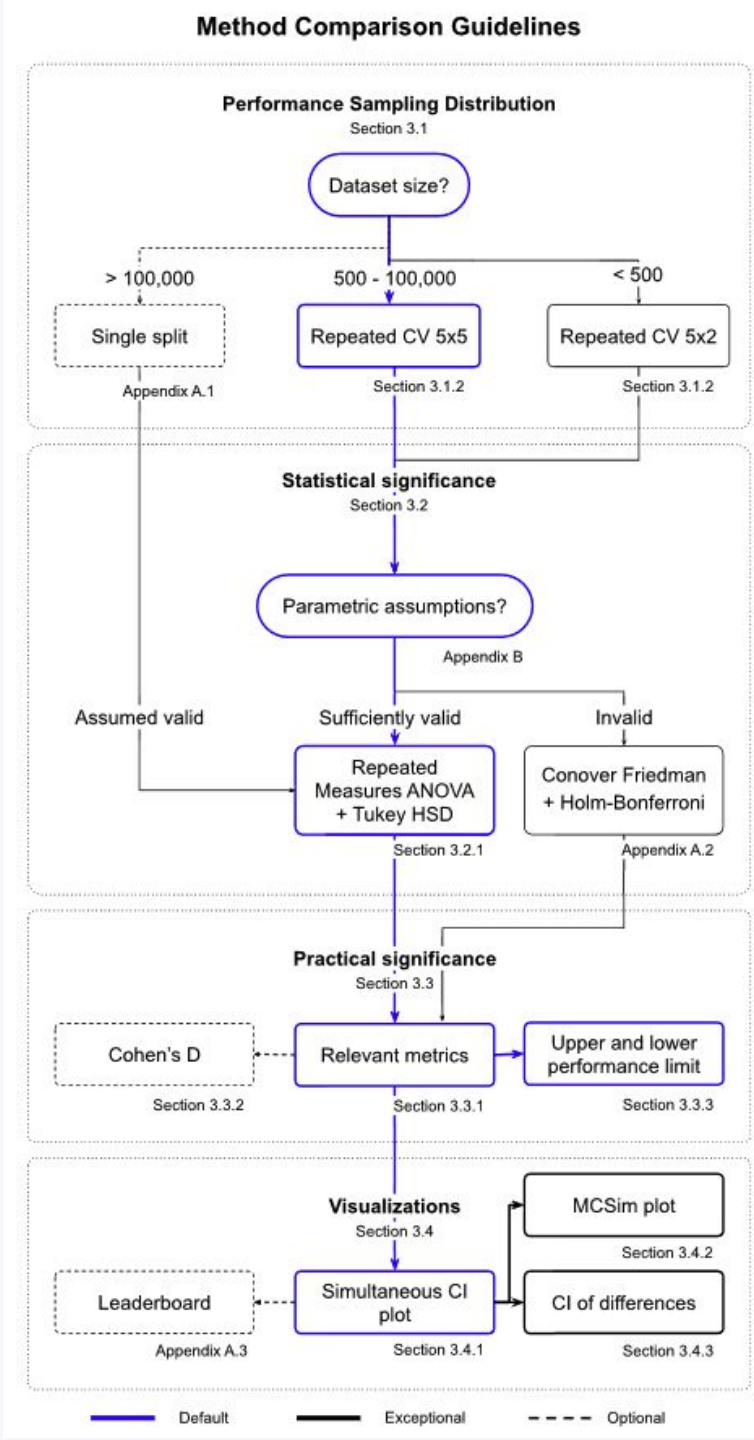
- Personal Motivation
- Phase 1: Training and Optimization
 - Generate informative baseline models
 - Optimize machine learning modelling
- Phase 2: Exploitation
 - Analyze the results from phase 1 unblinding and adapt the modelling
 - Generate final predictions
- Reflections
 - Surprises, regrets and suggested changes for future challenges

Why enter a blind challenge?

- Recently started as independent consultant
- My main goals:
 - Demonstrate how I would tackle a similar task
 - Generate an educational resource for anyone starting in cheminformatics
 - Provide content to share in LinkedIn
 - First project using coding assistants (Claude Code)
- Publicly release progress and code throughout the challenge:
 - Github repo: https://github.com/adlvd/pxr_challenge
 - My own website: <https://www.delavega.ai/blog.html>
 - LinkedIn posts

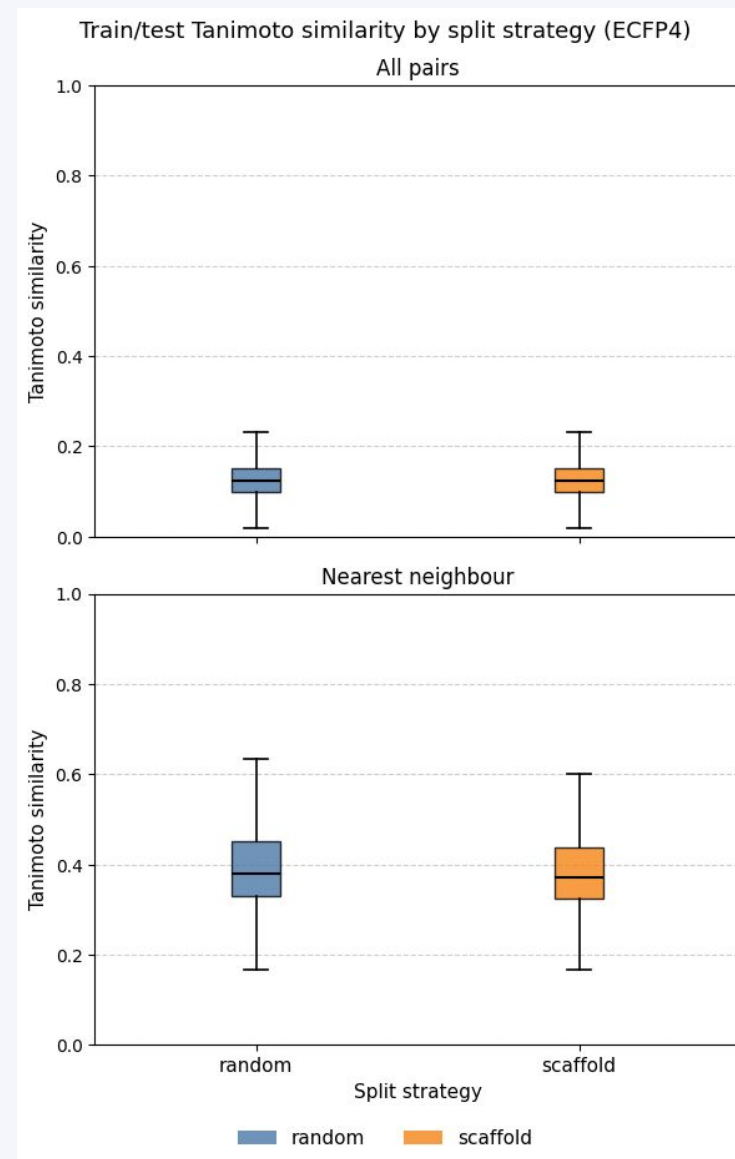
ML experimental design

- Statistically rigorous model comparison: 5x5 CV + repeated-measures testing (Ash et al., JCIM 2025)



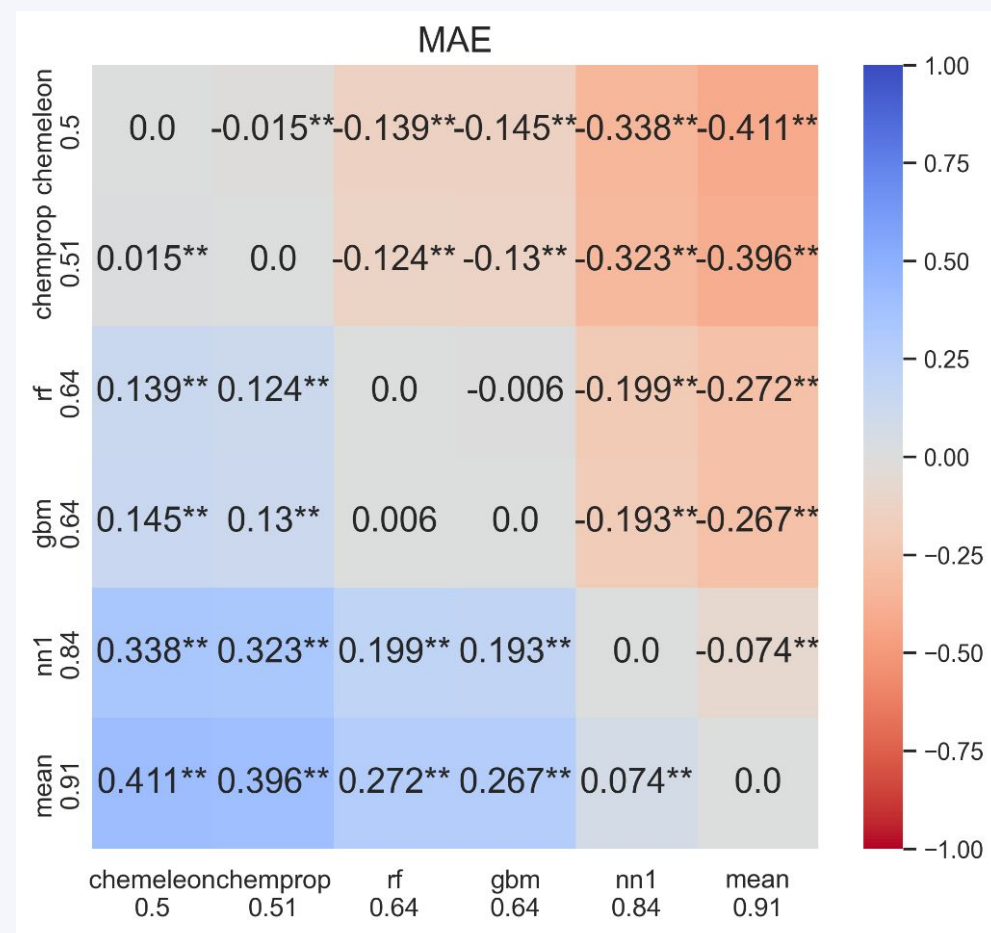
ML experimental design

- Statistically rigorous model comparison: 5x5 CV + repeated-measures testing (Ash et al., JCIM 2025)
- Potential data splits explored:
 - Random: chosen
 - Structure: not different from random
 - Temporal: not possible in this dataset
- Main performance metric: MAE



Generating first baseline models

- Models:
 - Chemprop
 - RF & GBT + ECFP4
 - CheMeleon (new model for me)
 - Mean (provides ceiling for MAE)
 - 1-Nearest Neighbors



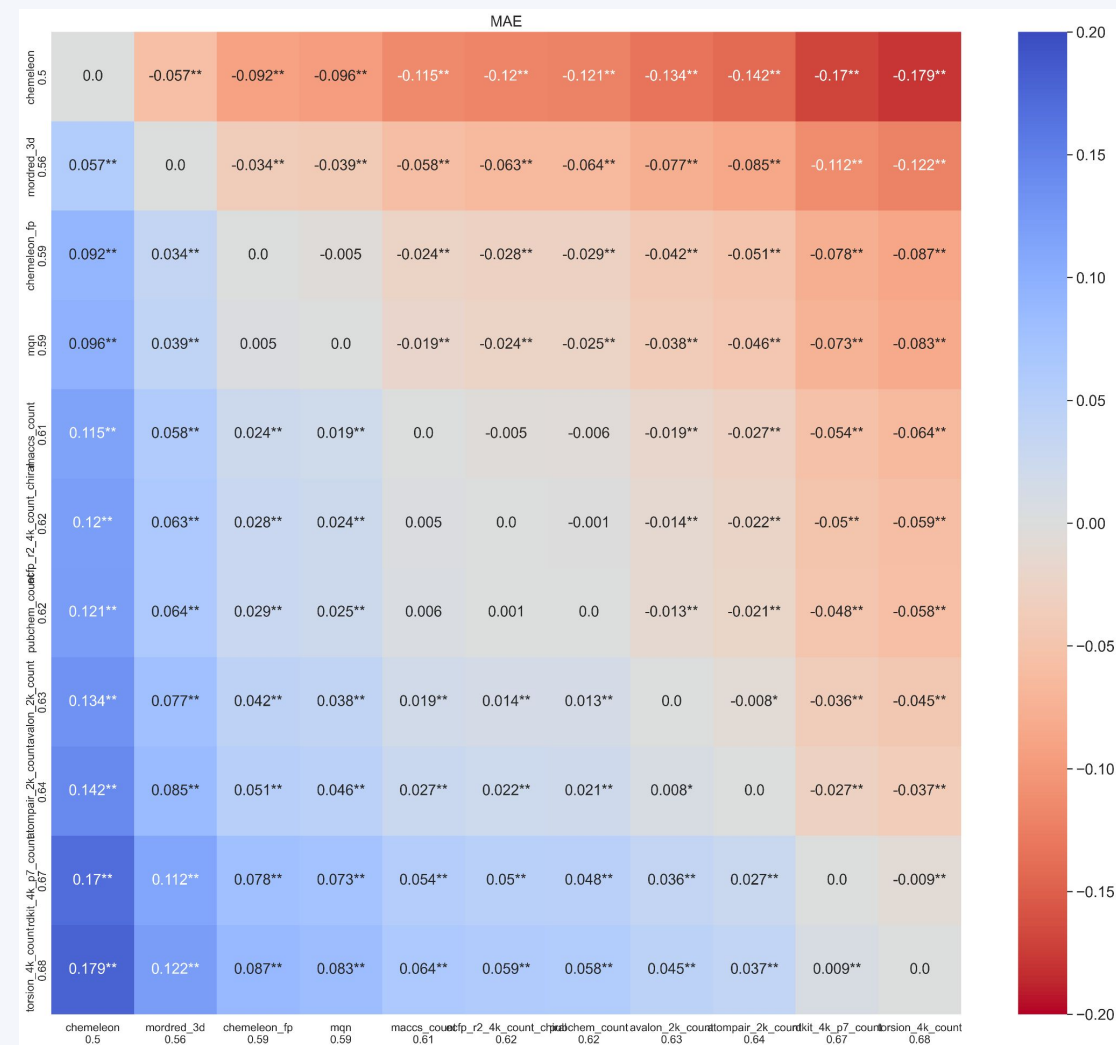
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- First submission to live leaderboard ended just above OpenADMET baseline

Model	Date	CV MAE	LB MAE	Rank
chameleon_base	22/04	0.500	0.574	57/84

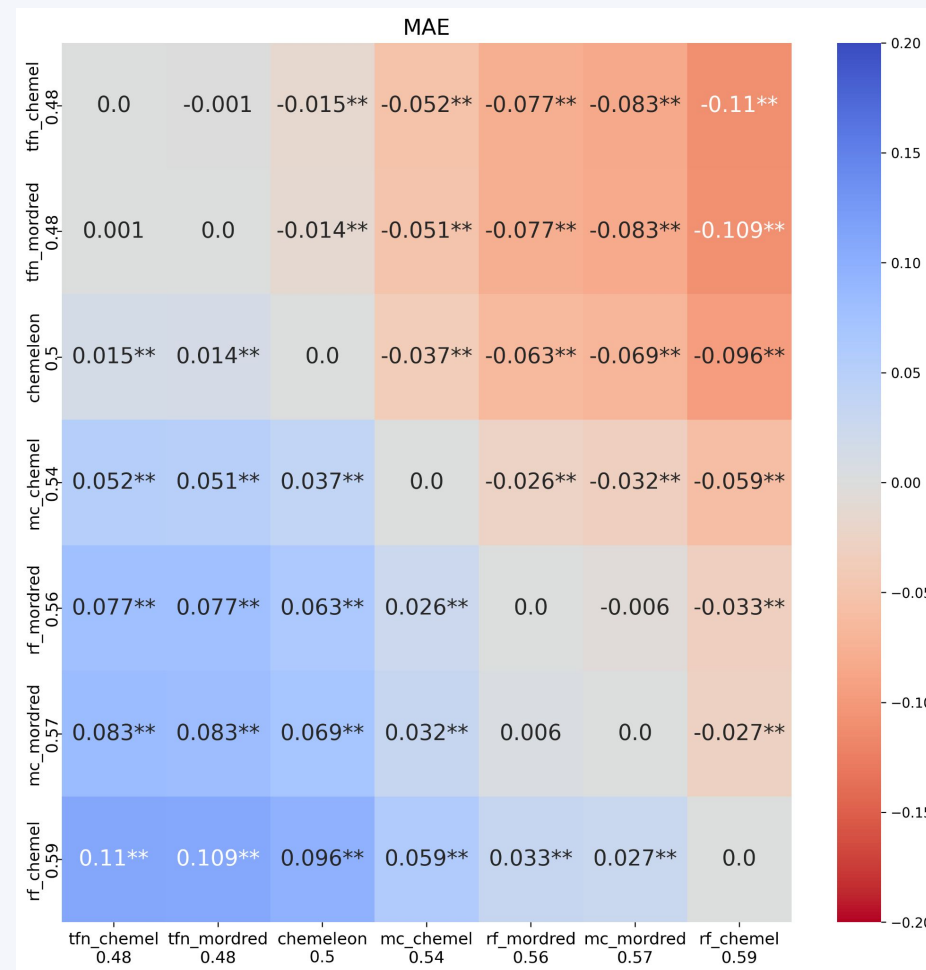
Improving the baselines

- Test large set of fingerprints on RF
 - Top3: Mordred, CheMeleon FP, MQN



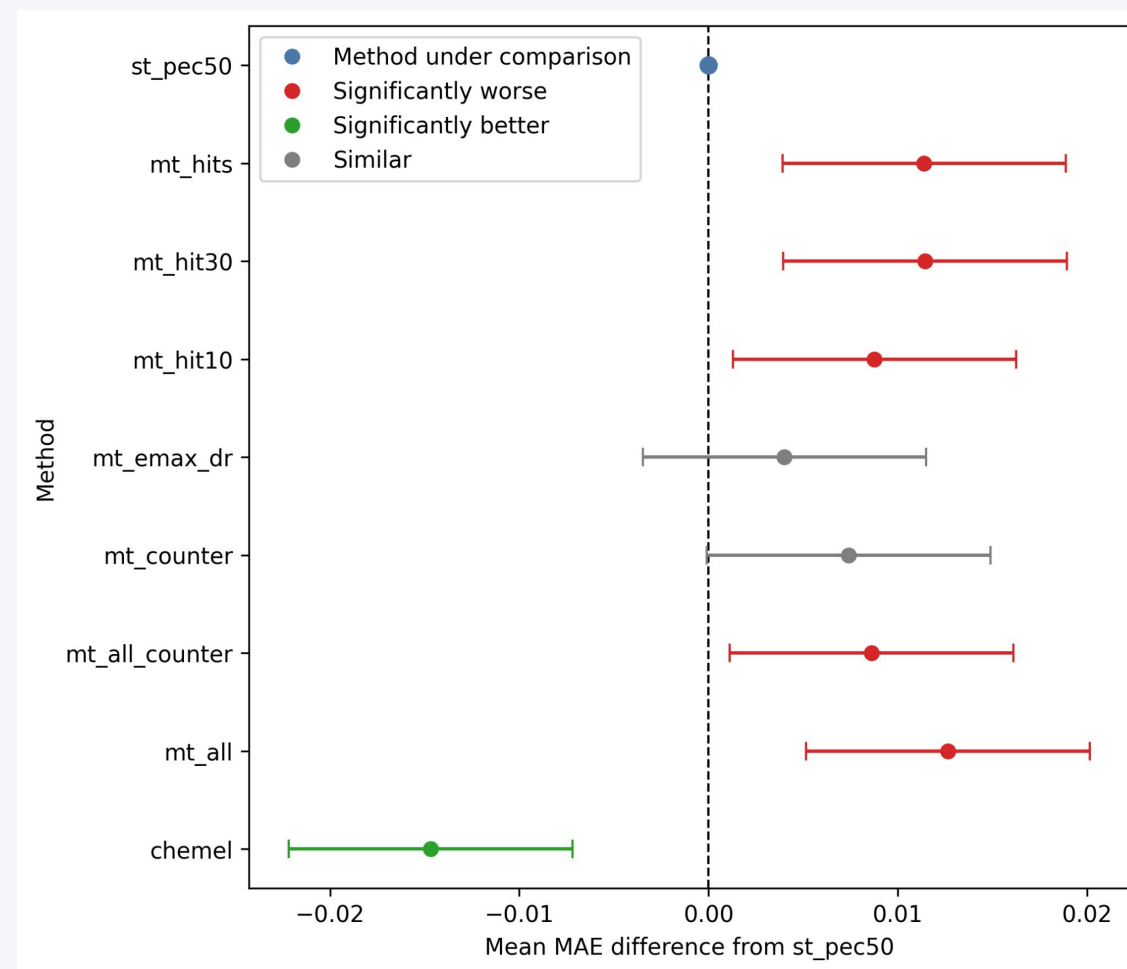
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- Test two new FP-based models:
 - Macau: matrix factorization technique
 - TabPFN: tabular pre-trained NN



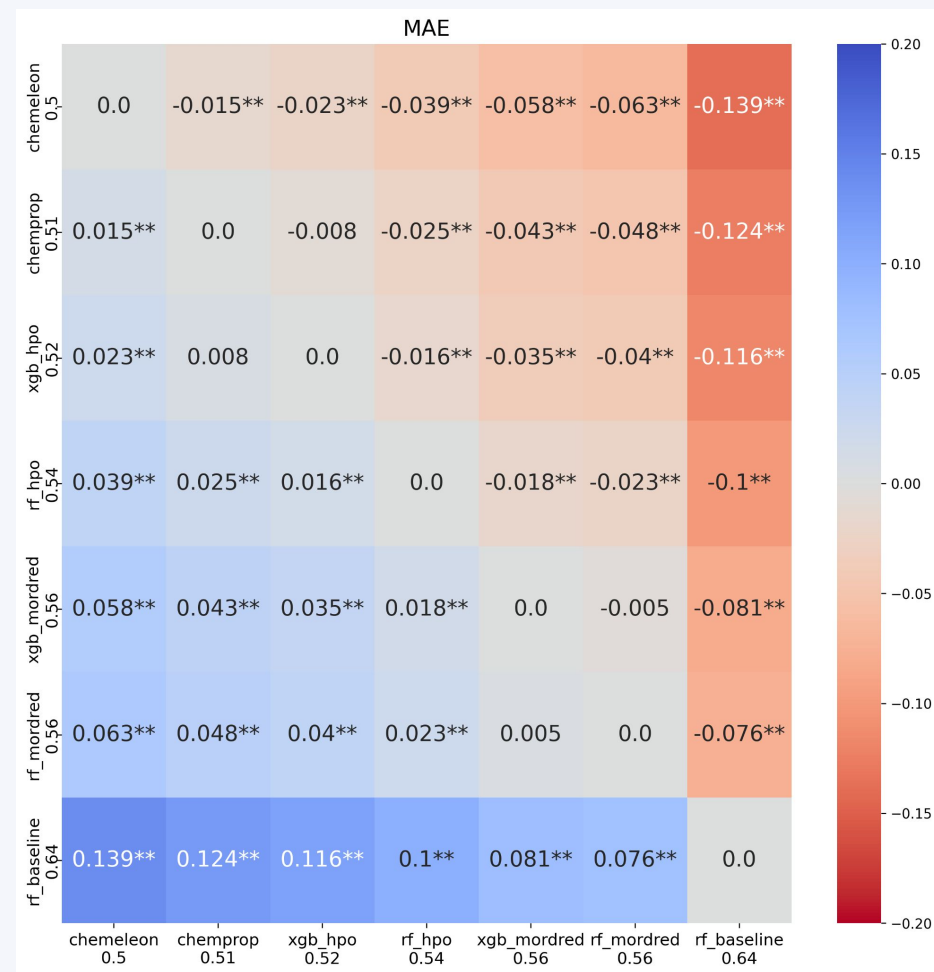
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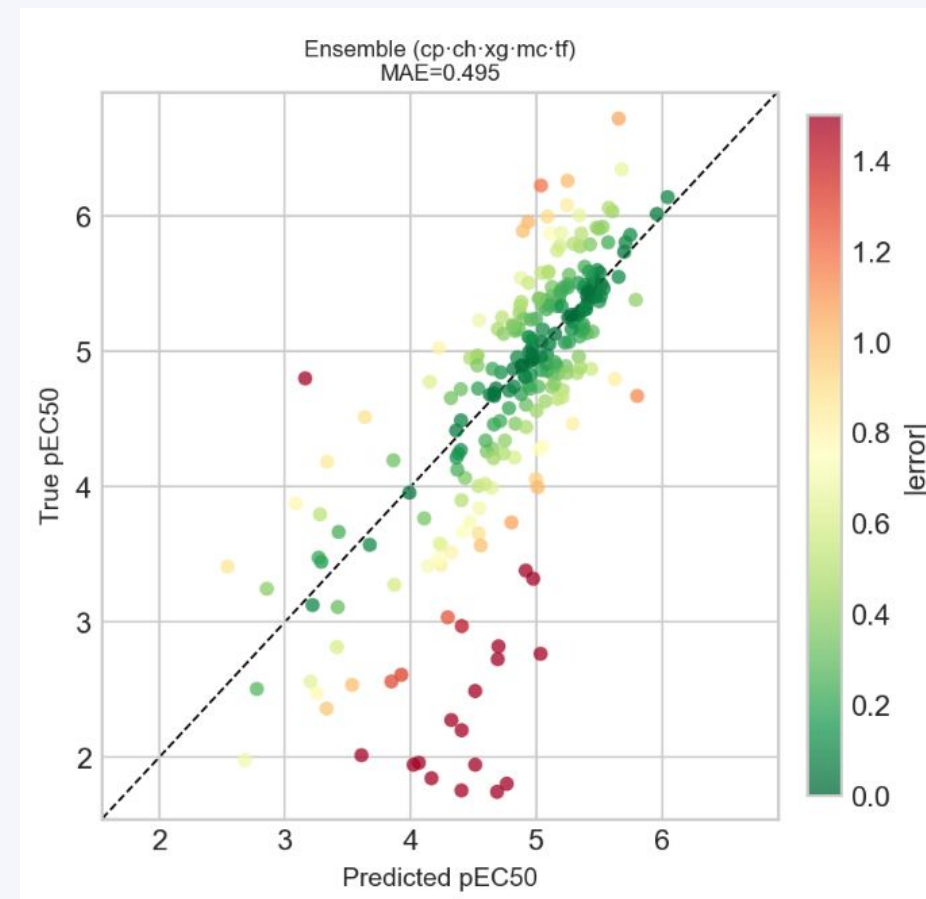
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- Ensembles based on weighted sums of prediction values

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rf0_cp1_ch2	29/04	0.483	0.521	65/156
rf1_cp1_ch2	30/04	0.493	0.515	61/156
rf1_cp2_ch2	30/04	0.490	0.507	51/156
rf0_cp1_ch1	30/04	0.484	0.503	50/159
P1 Ensemble	19/05	0.471	0.495	78/252

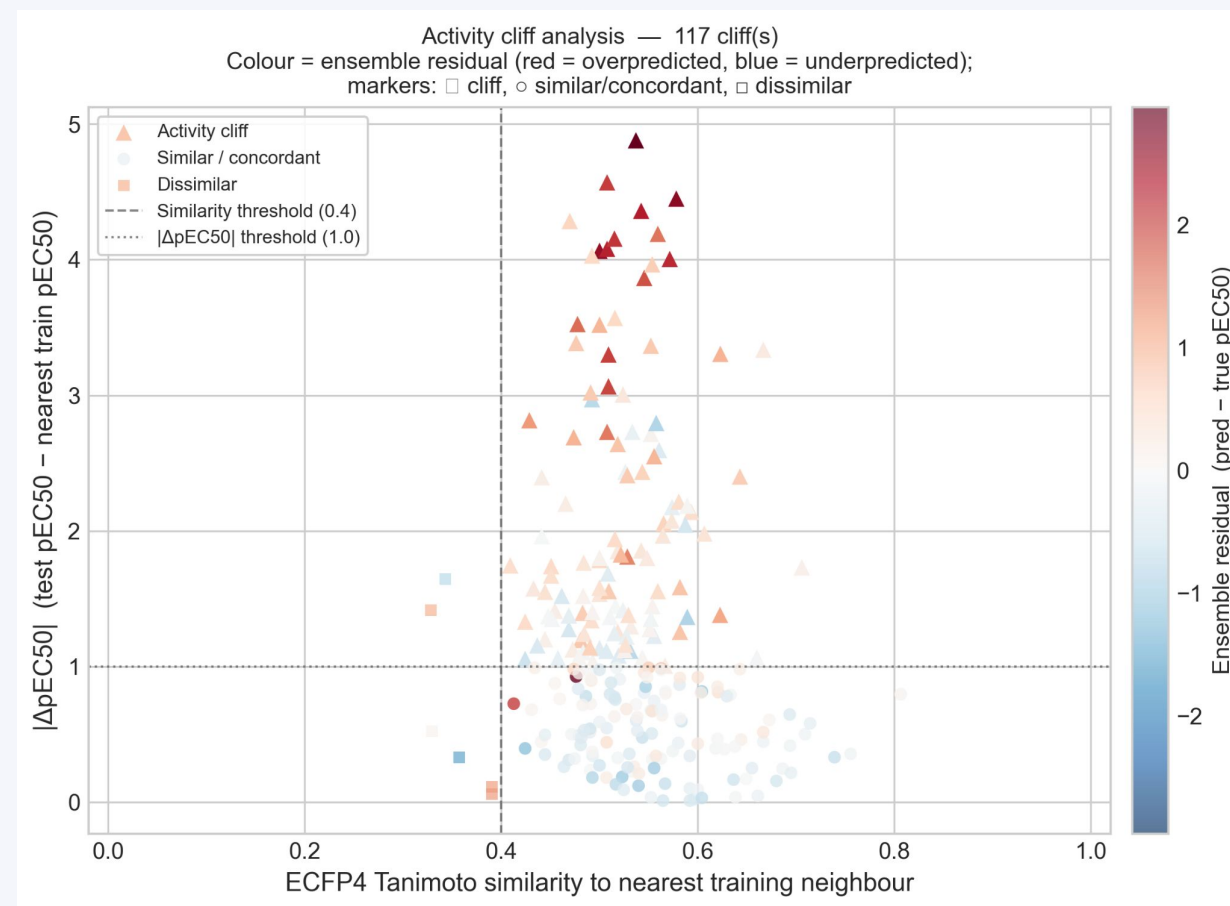
Unblinding the Phase 1 test data

- My MAE on the interim LB was 0.470 and my rank was 83 of 338
- Largest errors are found in low-activity compounds



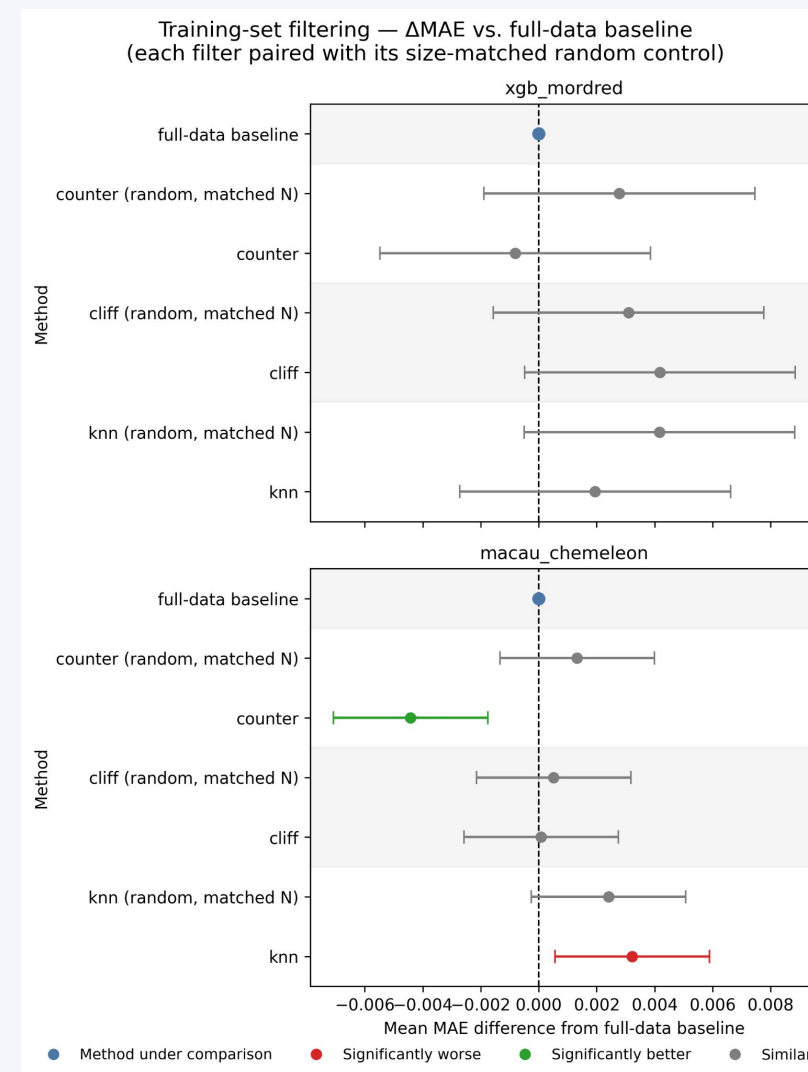
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- Activity cliffs to train compounds show largest errors in prediction



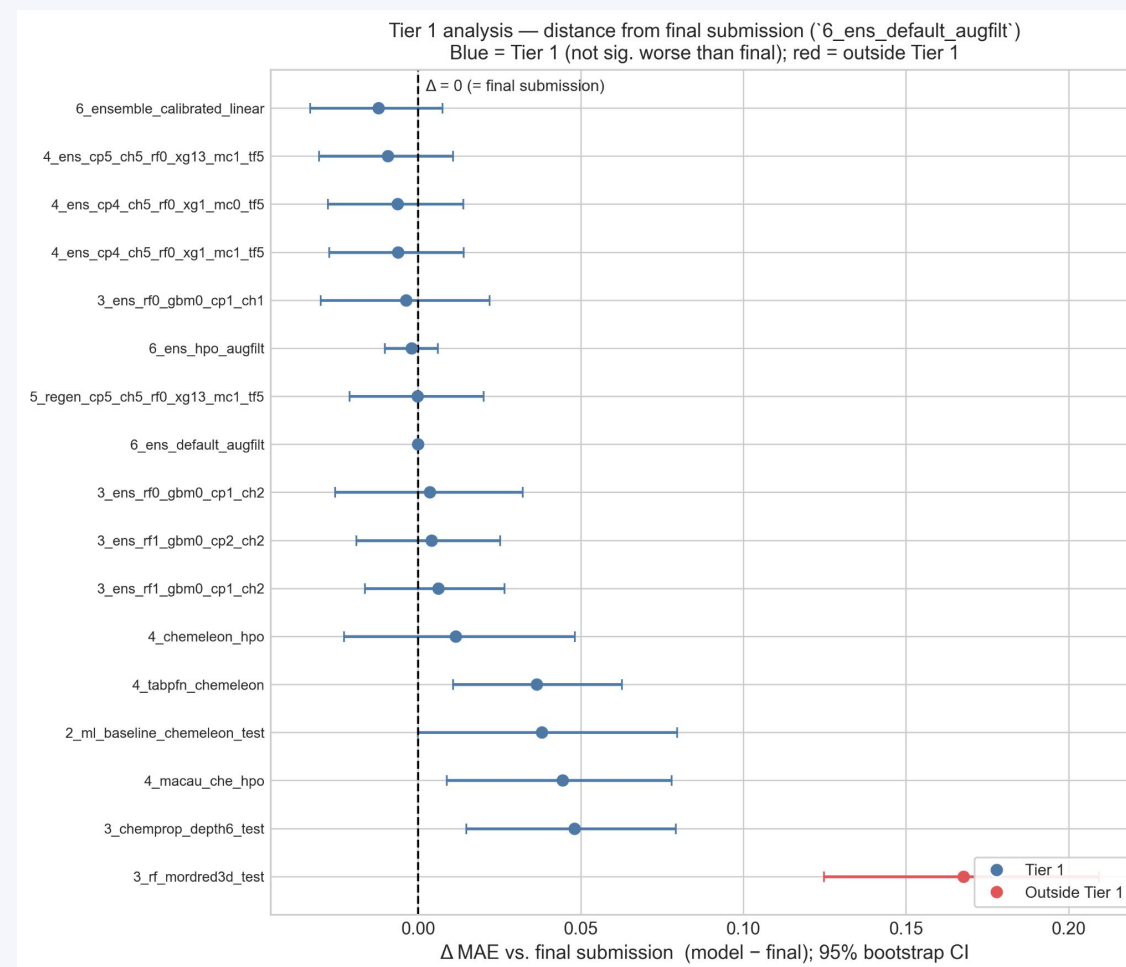
Final optimization round before end of phase 2

- Failed: Fix the bias in predictions
 - Post-hoc calibration
 - Sample weighting
 - Oversampling the extremes
- Partial success: Improve the training data
 - Add the semi-pure data
 - Filter the training data
- Final submission is a similar ensemble as in Phase 1 but with the improvements to the training data



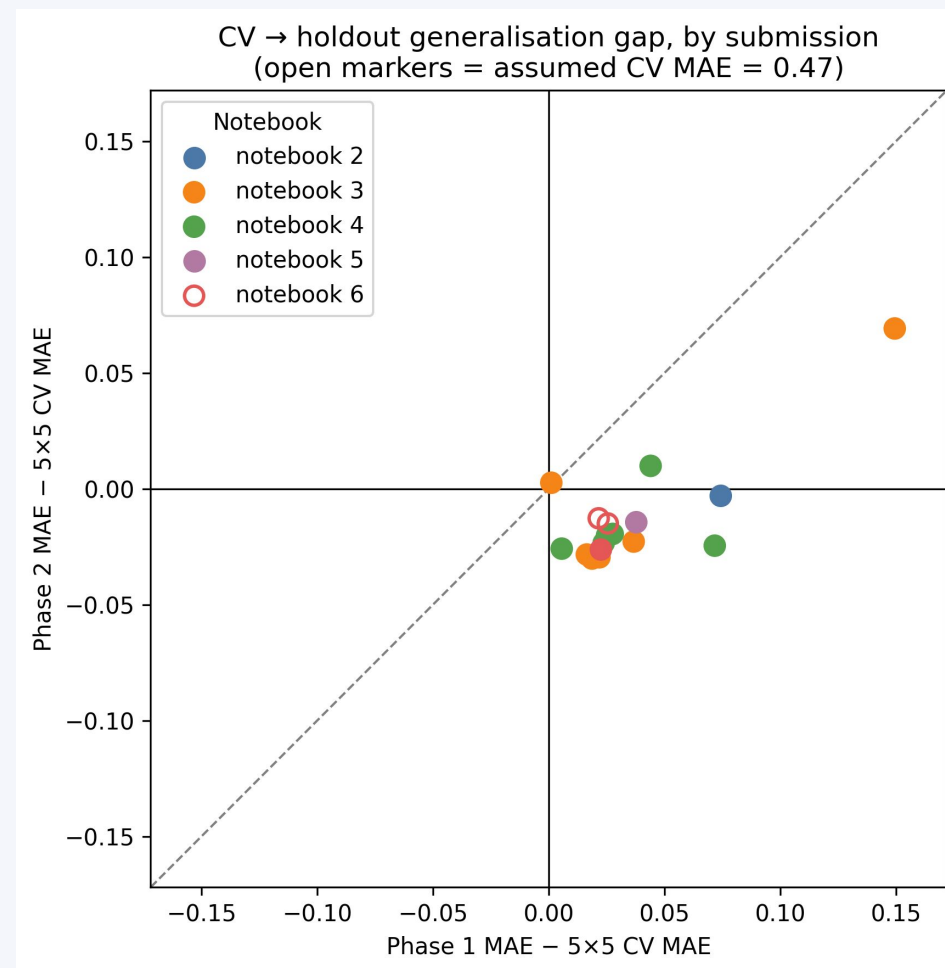
Submission results

- My final submission had MAE of 0.457 and rank 46 of 103 (Tier 2)
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- Across all submissions I generated, MAE values for phase 2 compounds were always better than for phase 1 compounds



My own takeaways

- My main regret:
 - Not making better use of the additional assay data provided and not finding time to look for and add public data to the analysis
- My biggest surprises:
 - How difficult it can be to compare submissions in a statistical manner
 - How much I was able to achieve with a relatively modest machine (Mac Mini M4 16GB)
- Did I achieve my goals? YES
 - I generated a useful resource for cheminformatics and drug discovery
 - I enjoyed the conversations and discussions in Discord

| My suggestions for future challenges

- Keep the two phase format and the live leaderboard in phase 1, but reduce the frequency/number of submissions allowed
- For deadlines, provide both date and time (with explicit time zones)

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- For deadlines, provide both date and time (with explicit time zones)
- Go from measuring model performance to measuring decision outcomes
 - E.g: rather than measure PXR prediction, highlight which compounds in the test set should be deprioritized due to likely PXR activation
 - **Final goal:** compare computational approaches to guide/optimize DMTL cycles

Thank you to the organizers and all other participants!

Links below if you want more info or to see the code

delavega.ai/blog

github.com/adlvd1/pxr_challenge

<https://www.linkedin.com/in/adlvd1/>

ML & cheminformatics consulting